

L1.7a Implicit Differentiation

Recall: L1.6 Trig Derivatives & the Chain Rule

Name _____

Solo – 9 minutes

1. Differentiate.

(a) $y = (3x^2 + 1)^5$

(b) $y = \cos(5x)$

2. Name the outer and the inner function. Do not differentiate anything.

	outer	inner
$\sqrt{\tan x}$		
$\tan(\sqrt{x})$		
$\sin^3 x$		

3. (a) Differentiate $y = x^2 \tan x$.

(b) The point $(3, 4)$ sits on the circle $x^2 + y^2 = 25$. **Using no calculus at all**, find the slope of the tangent line there.

